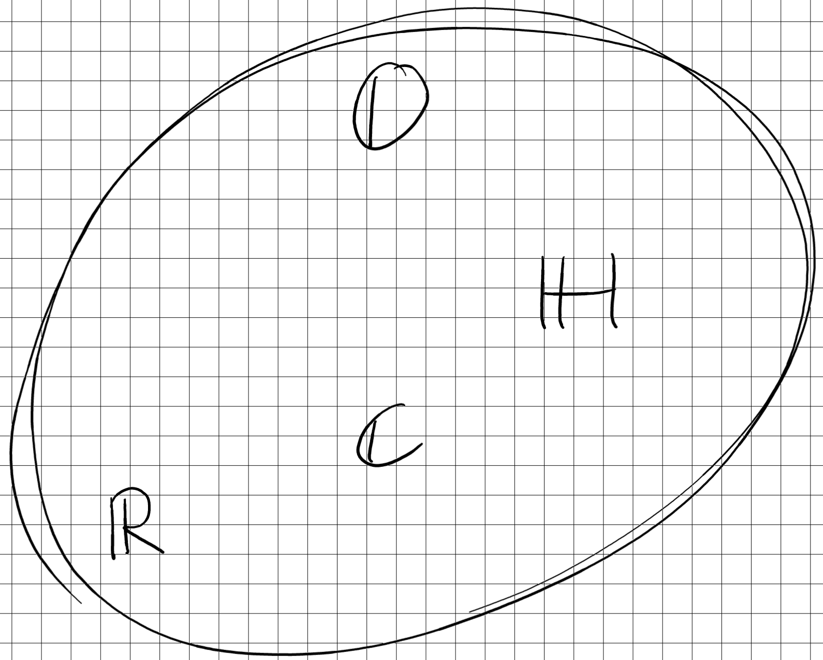



# Climbing Past the Complex Numbers

## the Cayley - Dickson Construction

or how to naturally build:

$\mathbb{R} \rightsquigarrow \mathbb{C} \rightsquigarrow \mathbb{H} \rightsquigarrow \mathbb{O} \rightsquigarrow \mathbb{S} \rightsquigarrow \dots$

under everything each of these is equipped with a conjugation $\alpha \mapsto \alpha^*$ such that

(1) $\alpha\alpha^* \in \mathbb{R}_0^+$ (this is a norm: $|\alpha|^2 := \alpha \cdot \alpha^*$

Note:

- complex numbers $\mathbb{C}$, is a field
  commutative, associative, inverses
- quaternions $\mathbb{H}$, division ring
  not commutativ, associative, inverses
- octonions $\mathbb{O}$
  not commutativ, not associativ, inverses
  but alternative: $x(xy) = (xx)y$ and $(xy)y = x(yy)$
- sedonions $\mathbb{S}$, has zero divisors
  nonassociative, inverses
  but power associative: $x(x(xx)) = (x(xx))x = (xx)(xx)$

note more on conjugation ...

in $\mathbb{R}$: $a \cdot a^* = a^2$

in $\mathbb{C}$: $(a+bi)(a-bi) = a^2+b^2 \geq 0$

in $\mathbb{H}$ : $(a+\underline{bi}+cj+dk)(a-\underline{bi}-cj-dk) = a^2+b^2+c^2+d^2$

$$\cancel{-bc\underbrace{ij}_{k}} \cancel{- bc\underbrace{ji}_{-k}}$$

in $\mathbb{O}$ : $(a+b_1e_1+\dots+b_7e_7)(a-b_1e_1-\dots-b_7e_7)$

$$= a^2+b_1^2+\dots+b_7^2 \geq 0$$

more properties of conjucation : ② $\alpha+\alpha^* \in \mathbb{R}$

③ $\alpha^{-1} = \frac{1}{|\alpha|^2}\cdot\alpha^*$

④ $(\alpha\beta)^* = \beta^*\alpha^*$

to the construction :

start with any algebra $A$ in the sequence and form $A' = A\times A$ with

- conjugation $(a,b)^* = (a^*,-b)$
- multiplication $(a,b)(c,d) = (ac-d^*b, da+bc^*)$

example $\mathbb{R} \rightsquigarrow \mathbb{C}$

$\mathbb{R}' = \mathbb{R}\times\mathbb{R}$

- $(a,b)^* = (a,-b)$
- $(a,b)(c,d) = (ac-bd, ad+bc)$

we have multiplicative identity $(1,0)(a,b) = (a,b)$

we have commutativity $(a,b)(c,d) = (ac-bd, ad+bc)$

$$= (ca-db, da+cb) = (c,d)(a,b)$$

we have a $\sqrt{-1}$ $(0,1)(0,1) = (-1,0)$

Now we show that this construction is isomorphic to $\mathbb{C}$ :

$\varphi: \mathbb{C} \to \mathbb{R}'$ , $a+bi \mapsto (a,b)$

observe $\varphi((a,b)(c,d)) = \varphi((ac-bd)+(ad+bc)i)$

$= (ac-bd, ad+bc) = (a,b)(c,d) = \varphi(a+bi)\cdot\varphi(c+di)$

$\Rightarrow \varphi$ preserves multiplication $\square$

example $\mathbb{C} \rightsquigarrow \mathbb{H}$

$\mathbb{C}' = \mathbb{C}\times\mathbb{C}$ , $(a,b)(c,d) = (ac-bd^*, ad+bc^*)$

$$(u+iv, x+iy)^* = (u-iv, -x-iy)$$

associative : $((a,b)(c,d))(e,f) = (ac-bd^*, ad+bc^*)(e,f)$

$= (ace-bd^*e-adf^*-bc^*f^*, acf-bd^*f+ade^*+bc^*e^*)$

$= (a,b)(ce-df^*, cf+de^*) = (a,b)((c,d)(e,f))$

has 3 square roots of $-1$ (up to negatives)

$(0,1)(0,1) = (-1,0)$ , $(i,0)(i,0) = (-1,0)$

and $(0,i)(0,i) = (-1,0)$ $\quad (*)\ \mathbb{1} = (1,0)$, $I = (i,0)$

$J = (0,1)$, $K = (0,i)$

note: $I^2 = J^2 = K^2 = -\mathbb{1}$

$IJK = -\mathbb{1}$

note: $C' = \operatorname{span}_{\mathbb{R}} \{\mathbb{1}, I, J, K\} \cong \mathbb{H}$

$$(a\mathbb{1} + bI + cJ + dK)^* = (a+bi, c+di)^* = (a-bi, -c-di)$$
$$= a\cdot\mathbb{1} - bI - cJ - dK$$

$\boxed{\text{example}}$ $\mathbb{H} \rightsquigarrow \mathbb{O}$

$\mathbb{H}' = \mathbb{H} \times \mathbb{H}$

not associative: $\big((0,i)(j,0)\big)(k,0) = (0,-k)(k,0) = (0,-1)$

$(0,i)\big((j,0)(k,0)\big) = (0,i)(i,0) = (1,0)$ $\quad \#$

$\mathbb{1} = (1,0)$ $\quad e_4 := e_1 \cdot e_2 = (0,1)(i,0) = (0,i)$

$e_1 := (0,1)$

$e_2 := (i,0)$ $\quad e_3 := (j,0)$

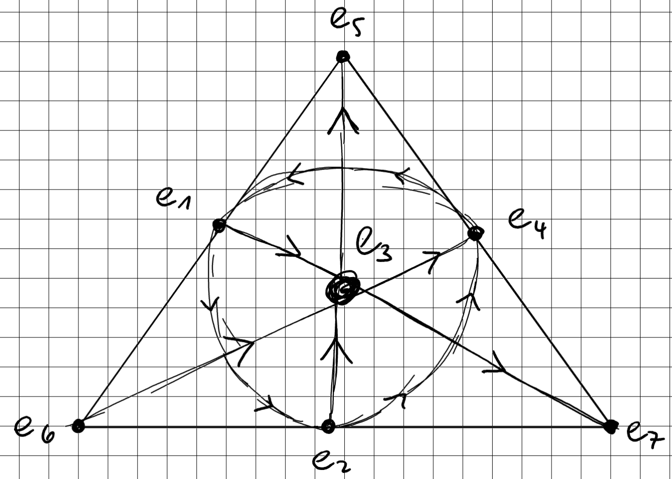
$e_5 := e_2 e_3 = (i,0)(j,0) = (k,0)$

$e_7 := e_1 e_3 = (0,1)(j,0) = (0,-j)$

$e_6 := e_1 e_5 = (0,1)(k,0) = (0,-k)$

also: $e_7 e_4 = (0,-j)(0,-i) = (k,0) = e_5$

visualization:




$e_4 e_3 = -e_6$

$e_2 e_6 = e_7$

$e_i^2 = -1$